15/02/24 FN

Hall Ticket Number:

MR-21

Code: B314CS5S

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. III Semester Supplementary Examinations, Jan/Feb-2024

Discrete Mathematics

(Common to CSE, CSM & CSD)

Time: 3 Hours

Max. Marks: 70

Answer ALL Questions

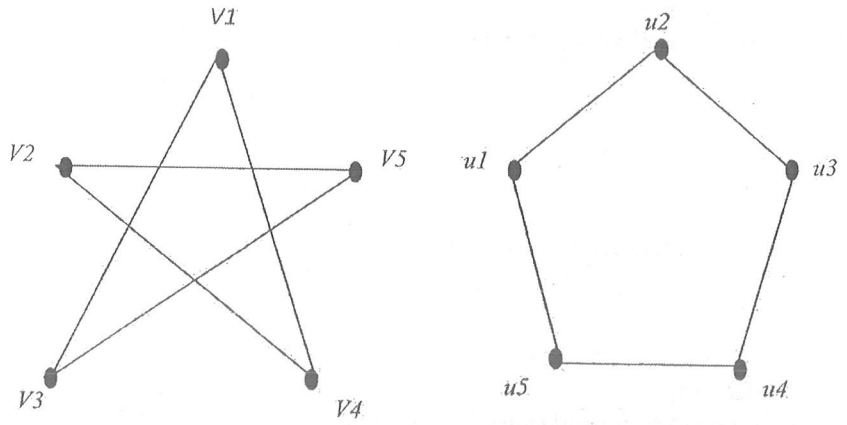
Part-A (10 x 2 = 20 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	What are the contrapositive, the converse, and the inverse of the conditional statement "The home team wins whenever it is raining".	2	1	1	1
1. b)	Give a direct proof of the theorem "If n is an odd integer, then n^2 is odd."	2	5	1	1
1. c)	Determine whether the function $f(x) = x + 1$ from the set of real numbers to itself is one-to-one.	2	5	2	1
1. d)	State and define any four properties of relation.	2	1	2	1
1. e)	Write a short note on well-ordering property.	2	1	3	1
1. f)	What is the expansion of $(x + y)^4$?	2	1	3	1
1. g)	Find the probability that a hand of five cards in poker contains four cards of one kind.	2	1	4	1
1. h)	List out applications of recurrence relations.	2	1	4	1
1. i)	What is complete graph and give any two examples.	2	1	5	1
1. j)	Elaborate the properties of trees.	2	1	5	1

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	Show that these conditional statements is a tautology by using truth tables: i. $[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$ ii. $[\neg p \wedge (p \vee q)] \rightarrow q$.	5	2	1	1
2. b)	Test that whether the given set of formulas are inconsistent $E \rightarrow S$, $S \rightarrow H$, $A \rightarrow \neg H$, $E \wedge A$.	5	3	1	1
	OR				
2. c)	Prove that if n is an integer and $3n+2$ is even, then n is even using a proof by contradiction.	5	2	1	1
2. d)	Show that $\forall x P(x) \vee \forall x Q(x)$ and $\forall x(P(x) \vee Q(x))$ are not logically equivalent.	5	2	1	1
3. a)	Find the inverse function of $f(x) = x^3 + 1$.	5	1	2	1
3. b)	Draw the Hasse diagram representing the partial ordering $\{(a/b) \mid a$ divides $b\}$ on $\{1,2,3,4,6,8,12\}$.	5	2	2	1
	OR				
3. c)	Let A = {a, b, c, d, e} and B = {a, b, c, d, e, f, g, h}. Find i) A ∪ B. ii) A ∩ B. iii) A − B. iv) B − A.	5	2	2	1
3. d)	Given that A ={1,2,3,4} and R ={(1,2),(2,3),(3,4)} be a relation on A. Find R^+.	5	3	2	1
4. a)	Give a recursion definition of the sequence $\{a_n\}$, n = 1,2,3,.... If a. $a_n = 6n$ b. $a_n = 2n + 1$ c. $a_n = 10^n$	5	3	3	1
4. b)	How many positive integers less than 1000 a. are divisible by 7? b. are divisible by 7 but not by 11? c. are divisible by both 7 and 11? d. are divisible by either 7 or 11?	5	2	3	1

P.T.O.

	OR				
4. c)	Construct a recursive version of a binary search algorithm.	5	3	3	1
4. d)	The English alphabet contains 21 consonants and five vowels. How many strings of six lowercase letters of the English alphabet contain a) Exactly one vowel? b) Exactly two vowels? c) At least one vowel? d) At least two vowels?	5	1	3	1
5. a)	Explain Bayes' theorem with an example.	5	2	4	1
5. b)	Find the solution to the recurrence relation $a_n = 6a_{n-1} - 11a_{n-2} + 6a_{n-3}$ with the initial conditions $a_0 = 2$, $a_1 = 5$, and $a_2 = 15$.	5	2	4	1
	OR				
5. c)	What is the variance of the number of times a 6 appears when a fair die is rolled 10 times?	5	1	4	1
5. d)	Find the coefficient of x^9 in the power series of $(x^3 + x^5 + x^6)(x^3 + x^4)(x + x^2 + x^3 + x^4 + \cdots)$ using Recurrence Relation.	5	2	4	1
6. a)	Determine whether the graphs are isomorphic.  	5	5	5	1
6. b)	Illustrate how Kruskal's algorithm and Prim's algorithm are used to find a minimum spanning tree, using a weighted graph with at least eight vertices and 15 edges.	5	2	5	1
	OR				
6. c)	What is Euler's formula? Illustrate with an example.	5	1	5	1
6. d)	Explain how to use breadth-first search to find the length of a shortest path between two vertices in an undirected graph.	5	2	5	1

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome